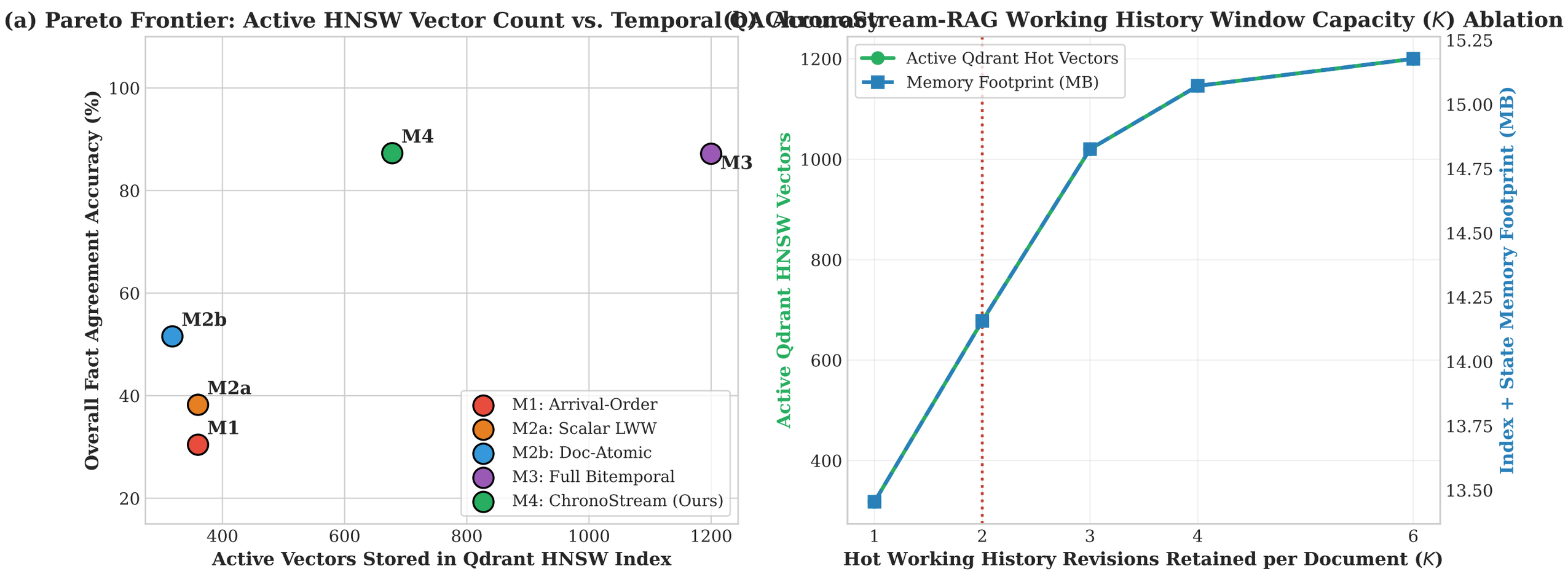
(a) Pareto Frontier: Active HNSW Vector Count vs. Temporal QA Accuracy
Overall Fact Agreement Accuracy (%)
Active Vectors Stored in Qdrant HNSW Index
M4
M3
M2b
M2a
M1
M1: Arrival-Order
M2a: Scalar LWW
M2b: Doc-Atomic
M3: Full Bitemporal
M4: ChronoStream (Ours)
(b) ChronoStream-RAG Working History Window Capacity (K) Ablation
Active Qdrant HNSW Vectors
Index + State Memory Footprint (MB)
Hot Working History Revisions Retained per Document (K)
Active Qdrant Hot Vectors
Memory Footprint (MB)